Semantic Restrictions on Certain Complementizers

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Any researcher interested in the relation between linguistic form and its semantic content will find quite natural the question of whether or not the absence of some particular linguistic form can be explained on semantic grounds. In this note I will show that the impossibility of a *wh*-complementizer after *neg*-transportation verbs has semantic justification.

Formally, one defines *neg*-transportation verbs as verbs taking two arguments, a nominal one and a sentential one, such that negation of the verb itself can be "transported" to the sentential argument without important semantic change. More precisely a *neg*-transportation verb forms, with its subject NP, a sentential operator O such that "not-O(P)" implies "O(not-P)" (where 'P' is the sentential argument of the *neg*-transportation verb).

Thus, in the following examples, (1) implies (2) and (3) implies (4):

(1) Bill does not believe that Sue will call
(2) Bill believes that Sue will not call
(3) It does not seem that it will rain
(4) It seems that it will not rain

There exists a well-known objection to the claim that (1) implies (2). According to this objection (1) cannot imply (2) since there can be situations, when Bill does not exist or when Bill does not know Sue, in which (1) is true and (2) is false. This objection raises the problem of specifying the kind of the negation used in (1) and (3). Without entering into details I can affirm that this objection can be avoided if we consider that the negation in question is *opacity preserving*. *Neg*-transportation verbs are propositional attitude verbs (or are reducible to them) and as such they are (strongly) opaque or intensional. It means that the replacement of an argument-expression under the scope of the *neg*-transportation verb by another argument-expression which is extensionally equivalent to the replaced one, can provoke the change of the truth value of the whole complex construction. Thus, in the following examples, (5) and (6) can have different truth values even if *Sue* and *John's sister* refer to the same person:

(5) Bill thinks that Sue will call
(6) Bill thinks that John's sister will call

In fact verbs of propositional attitude are very strongly opaque; their opacity can be detected, as opposed to that of classical modal operators, by contingent sentences. More precisely "O" is strongly opaque iff, for every contingent sentence P and for every situation s, if O(P) is true at s, then there exists a contingent sentence P′ with the same truth value as P at s and such that O(P′) is false at s. *Neg*-transportation verbs cannot take *wh*-complementizers: the following forms are impossible: **to think whether*, **to hope who*, **to believe why*, **to be afraid what*, etc. If these forms were possible they would have semantic properties similar to those with which such complementizers are possible. To obtain these properties we will consider the constructions with *to know whether* and semantic relations into which they enter. It can be noticed first of all that (7) and (8) both imply (9):

(7) Bill knows that Sue will call
(8) Bill knows that Sue will not call
(9) Bill knows whether Sue will call or not

More precisely we have the following property:

Prop. 1: "O that P" implies "O whether P or not"
"O that not-P" implies "O whether P or not"

The second property of *wh*-constructions we need says that the opacity preserving negation of *that*-construction implies the negation of the corresponding *wh*-construction: the opacity preserving negation of (7) and of (8) implies the opacity preserving negation of (9). For instance (10) implies (11):

(10) Bill does not know that Sue will call
(11) Bill does not know whether Sue will call or not

This property can be stated in general in the following way:

Prop. 2: If "O(P)" implies "O′(P)" and "O" and "O′" are operators of the same opacity strength, then "not-O(P)" implies "not-O′(P)"

Two operators have the same opacity strength if any pair of equivalent sentences which can be used to detect the opacity of the first operator can also be used to detect the opacity of the second one. For instance the operator formed from *to regret that* is not of the same strength as the operator formed from *to know that* because the pair (12) and (13) of equivalent sentences does not behave in the same way with respect to those two operators:

(12) The bottle is half full
(13) The bottle is half empty

Indeed, only (15a) and (15b) can have different truth values; (14a) and (14b) have the same truth values:

(14a) Bill knows that the bottle is half full
(14b) Bill knows that the bottle is half empty
(15a) Bill regrets that the bottle is half full
(15b) Bill regrets that the bottle is half empty

Thus the pair (12) and (13) shows the opacity of *to regret* and not the opacity of *to know that* and consequently these two operators are not of the same strength. On the other hand, as one can easily verify, *to know that* and *to know whether* are of the same opacity strength.

Now we can demonstrate the property Prop. 2. Suppose that "O(P)" implies O′(P) but that "not-O(P)" does not imply "not-O′(P)". There then exists a situation *s* in which "not-O(P)" is true and "not-O′(P)" is false. But since *not* is an opacity preserving negation, *not-O* and *not-O′* are new opaque operators of the same strength. Consequently there exists a sentence P′ with the same truth value as P at *s* such that "not-O(P′)" is false and "not-O′(P′)" is true at *s*. But then, because of the usual semantics of the negation, "O(P′)" is true and "O′(P′)" is false. But this cannot be possible since "O(P′)" implies "O′(P′)". So the property Prop 2 must be true.

The property Prop 2 in conjunction with the definition of the *neg*-transportation verb entails the impossibility of having *wh*-complementizers with these verbs. To show that suppose that "O" is a *neg*-transportation verb. Then, according to the definition of such a verb we have (16):

(16) "not-O that P" implies "O that (not-P)"

Furthermore, if we had *O whether* form, then according to the property *Prop 1* (17) would be true:

(17) "O that not-P" implies "O whether P or not"

The conjunction of (16) and (17) gives (18):

(18) "not-O that P" implies "O whether P or not"

On the other hand the Prop 2 expresses what is given in (19):

(19) "not-O that P" implies "not-O whether P or not"

But (18) and (19) are in contradiction. So our supposition that there exists a "O whether" form, where "O" is a *neg*-transportation verb, leads to a contradiction.

Similar reasoning holds for other *wh*-complementizers. Instead of using the Prop 1 and Prop 2 which holds for *whether*-complementizer, one should generalise these properties using the well-known decomposition of other *wh*-quantifiers (such as *what, who, where,* etc.)

It seems important to point out in conclusion that what I have shown is not that *whether*-complementizers are absolutely impossible with *neg*-transportation verbs. I have shown that they are impossible only with their "main" meaning, the one given by the definition of *neg*-transportation verb and by the properties *Prop 1* and *Prop 2.* In particular they must be related to the *that*-complementizer in the way shown above. Consequently it may happen that in some languages one can find, in some construction, **to think whether* or its equivalent. My claim is however that in this case the meaning of the verb will be different from the one stipulated by the properties here discussed.